Hridey Marwah

Delhi, India | hrideymarwah2907@gmail.com | Portfolio | GitHub | LinkedIn

PROFILE

First-year Computer Science (AI) student researching and building AI systems, multi-agent architectures, and full-stack applications. Exploring reliable orchestration, agentic coding, memory, reasoning, tool use, and verification while contributing to open-source machine-learning infrastructure.

EDUCATION

Newton School of Technology **2025–2029**
B.Tech in Computer Science & Engineering (AI), Rishihood University, Sonipat

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, HTML, CSS **Frontend:** React, Next.js, Tailwind CSS
Backend/Data: Node.js, FastAPI, REST APIs, PostgreSQL, Prisma, Supabase **AI/ML:** JAX, NumPy, LLMs, Multi-Agent Systems, OCR, Computer Vision
Tools: Git, GitHub, VS Code, Docker, Ollama, MCP

PROJECTS

Brain Architecture — Cognitive AI Orchestration *Active research*
Agent orchestration / shared memory / experimental systems
- Researching how specialized agents, shared memory, planning, tool use, and failure recovery can coordinate on complex multi-step tasks.
- Designing critique, synthesis, execution, and verification loops that treat first-pass model output as an intermediate result.

Co-Team + AeroHarness — Multi-Agent Coding System *Co-Team* / *AeroHarness*
TypeScript / VS Code / MCP / SQLite / Ollama
- Built a VS Code multi-agent pipeline that classifies task complexity and routes work across specialized planning, coding, design, documentation, and review agents.
- Separated orchestration from controlled execution through a platform-agnostic harness with sandboxing, human approval gates, event traces, and iterative verification.

Supply Chain Disruption Simulator *Repository*
React / Node.js / PostgreSQL / Prisma / JavaScript
- Built an interactive event-driven simulation of disruptions propagating across directed supplier, warehouse, distribution, and retailer networks.
- Modeled cascading inventory, order, and shipment effects; compared alternate routes, suppliers, and recovery strategies.

RoleReady AI — Adaptive Career Roadmaps *Repository*
Python / FastAPI / Sentence Transformers / NetworkX
- Built a resume-to-job skill-gap pipeline with semantic matching, readiness scoring, and prerequisite-aware learning paths.
- Used a skill dependency graph and topological ordering to generate structured roadmaps with explainable recommendations.

Aushadhi OCR — Medical Prescription Digitization *Repository*
OCR / Computer Vision / Next.js / TypeScript / Supabase
- Built an OCR workflow that extracts medicine and prescription details into structured verification records.
- Applied image preprocessing and AI-assisted correction to improve extraction from noisy, unstructured medical text.

OPEN SOURCE

JAX Contributor 4 merged pull requests
- Contributed fixes to JAX numerical behavior and error handling, including boundary derivatives and safer invalid-code handling.

ACHIEVEMENTS

Hackathon Finalist — IIIT Gwalior InfoTsav'25 | **IISc Bangalore Hackathon** — Participant | **GSoC 2026 Applicant**